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Quantum Computation and Quantum Information Theory (QCQIT) (Spring 2026) Lecture Notes

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1 Lecture#1 (Jan 20)

[**Linear Algebra Review**]

[**Vector Spaces**] Throughout, all vector spaces are over the complex field $\mathbb{C}$

[**Notation.**] Vectors in a vector space V are denoted using Dirac notation, e.g., $|v\rangle \in V$.

[**Inner Product**] An inner product on a vector space V is a map $(\cdot, \cdot) : V \times V \rightarrow \mathbb{C}$ that satisfies the following properties:

1. $(|v\rangle, \sum_{i=1}^n c_i |w_i\rangle) = \sum_{i=1}^n c_i (|v\rangle, |w_i\rangle)$, where $c_i \in \mathbb{C}$
2. $(|v\rangle, |w\rangle) = (|w\rangle, |v\rangle)^*$
3. $(|v\rangle, |v\rangle) \geq 0$ (it is equal to 0 if and only if $|v\rangle = 0$)

We also write the inner product as $\langle v|w\rangle$. A inner product space is a vector space equipped with an inner product. The norm of a vector $|v\rangle \in V$ is defined by $\| |v\rangle \| = \sqrt{\langle v|v\rangle}$.

[**Exc**] Show that $(\sum_{i=1}^n c_i |v_i\rangle, |w\rangle) = \sum_{i=1}^n c_i^* (|v_i\rangle, |w\rangle)$

[**Linear Operator**] For vector spaces V, W , a map/operator $A : V \rightarrow W$ is called linear if

$$A(\alpha |v\rangle + \beta |w\rangle) = \alpha A(|v\rangle) + \beta A(|w\rangle)$$

for all $\alpha, \beta \in \mathbb{C}$ and $|v\rangle, |w\rangle \in V$.

[**Linear Operator vs Matrix Representation**] Let V and W be vector spaces with ordered bases $\{|v_i\rangle\}_{i=1}^n$ and $\{|w_j\rangle\}_{j=1}^m$, respectively. Let A be an $m \times n$ matrix with entries in $\mathbb{C}$. Define a map $T_A : V \rightarrow W$ as follows: for $|v\rangle \in V$ written as $|v\rangle = \sum_{i=1}^n \alpha_i |v_i\rangle$, set $A|v\rangle = \sum_{j=1}^m \beta_j |w_j\rangle$ where the coefficients β_j are given by

$$\begin{bmatrix} \beta_1 \\ \vdots \\ \beta_m \end{bmatrix} = A \cdot \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$$

Check that T_A is linear.

Conversely, given any linear operator $T : V \rightarrow W$, there exists an $m \times n$ matrix A over $\mathbb{C}$ (depending on the chosen bases of V and W) such that $T = T_A$.

[**The Pauli Matrices**] Four particularly important matrices that describe linear operators $A : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ are the Pauli matrices:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

[Dual: $\langle v|$] For any vector $|v\rangle \in V$, define the map $\langle v| : V \rightarrow \mathbb{C}$ as follows: for any $|w\rangle \in V$,

$$\langle v|(|w\rangle) = \langle v|w\rangle$$

Check that $\langle v|$ is a linear operator. Suppose $\{|i\rangle\}$ be an orthonormal basis of V . Then, the matrix representation of $\langle v|$ is given by: let $|v\rangle = \sum_i \alpha_i |i\rangle$. Take any $|w\rangle = \sum_j \beta_j |j\rangle \in V$. Then

$$\langle v|(|w\rangle) = \langle v|w\rangle = [\alpha_1^*, \dots, \alpha_n^*] \cdot \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}$$

Indeed,

$$\begin{aligned} \langle v|(|w\rangle) &= \langle v|w\rangle \\ &= \left(\sum_i \alpha_i |i\rangle \mid \sum_j \beta_j |j\rangle \right) \\ &= \sum_{i,j} \alpha_i^* \beta_j \langle i|, |j\rangle \rangle \\ &= \sum_i \alpha_i^* \beta_i \end{aligned}$$

[Outer Product] For $|v\rangle \in V$, $|w\rangle \in W$, define the *outer product* maps $|w\rangle\langle v| : V \rightarrow W$ as follows: for any $|v'\rangle \in V$

$$|w\rangle\langle v|(|v'\rangle) = |w\rangle \langle v|v'\rangle$$

Claim Outer product is linear.

[Completeness Result] Let $\{|i\rangle\}$ be an orthonormal basis of V . Then $\sum_i |i\rangle\langle i| = I_V$.

Proof: Take any $|v\rangle \in V$. Let $v = \sum_i \alpha_i |i\rangle$. Then

$$\begin{aligned} \left(\sum_i |i\rangle\langle i| \right) (|v\rangle) &= \sum_i (|i\rangle\langle i|)(|v\rangle) \\ &= \sum_i |i\rangle \langle i|v\rangle \\ &= \sum_i |i\rangle \alpha_i \\ &= |v\rangle \end{aligned}$$

[Fact] Let V and W be inner product spaces with $\{|v_i\rangle\}$ and $\{|w_j\rangle\}$ as orthonormal bases respectively. Let $A : V \rightarrow W$ be any linear operator. Then

$$A = \sum_{i,j} \langle w_j|A|v_i\rangle |w_j\rangle\langle v_i|$$

where $\langle w_j|A|v_i\rangle = \langle w_j|, A|v_i\rangle \in \mathbb{C}$.

Proof: We first write,

$$\begin{aligned}
A &= I_W A I_V \\
&= \left(\sum_j |w_j\rangle\langle w_j| \right) A \left(\sum_i |v_i\rangle\langle v_i| \right) \\
&= \sum_{i,j} (|w_j\rangle\langle w_j|) A (|v_i\rangle\langle v_i|)
\end{aligned}$$

For a fix i, j and for any $|v\rangle \in V$, we have

$$\begin{aligned}
(|w_j\rangle\langle w_j|) A (|v_i\rangle\langle v_i|) (|v\rangle) &= (|w_j\rangle\langle w_j|) A |v_i\rangle\langle v_i|v\rangle \\
&= (|w_j\rangle\langle w_j|) \langle v_i|v\rangle A |v_i\rangle \\
&= \langle v_i|v\rangle (|w_j\rangle\langle w_j|) A |v_i\rangle \\
&= \langle v_i|v\rangle |w_j\rangle\langle w_j| A |v_i\rangle \\
&= \langle w_j| A |v_i\rangle\langle v_i|v\rangle |w_j\rangle \\
&= \langle w_j| A |v_i\rangle |w_j\rangle\langle v_i| (|v\rangle)
\end{aligned}$$

Therefore, as a map, $|w_j\rangle\langle w_j| A (|v_i\rangle\langle v_i|) = \langle w_j| A |v_i\rangle |w_j\rangle\langle v_i|$. Hence,

$$A = \sum_{i,j} (|w_j\rangle\langle w_j|) A (|v_i\rangle\langle v_i|) = \sum_{i,j} \langle w_j| A |v_i\rangle |w_j\rangle\langle v_i|$$

2 Lecture#2 (Jan 23)

[Exc] Consider the linear operator $A_Z : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ defined by the matrix $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Let $\{|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}\}$ be the standard basis. Show that

$$A_Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

[The Cauchy-Schwarz Inequality] For any vectors $|v\rangle, |w\rangle \in V$,

$$|\langle v|w\rangle|^2 \leq \langle v|v\rangle\langle w|w\rangle$$

Proof:

[Adjoint] Consider the inner product spaces V and W . For any linear operator $A : V \rightarrow W$, the adjoint of A , denoted by $A^\dagger$, is the unique operator $A^\dagger : W \rightarrow V$ such that

$$\langle |v\rangle, A |w\rangle \rangle = \langle A^\dagger |v\rangle, |w\rangle \rangle, \text{ for all } |v\rangle \in V, \text{ and } |w\rangle \in W$$

[Exc] Show that $A^\dagger$ is unique and also it is linear.

[Exc] $\langle A |v\rangle, |w\rangle \rangle = \langle |v\rangle, A^\dagger |w\rangle \rangle$, for all $|v\rangle \in V$ and $|w\rangle \in V$

[Exc] $(A^\dagger)^\dagger = A$

[Exc] For linear operators A, B , show that $(AB)^\dagger = B^\dagger A^\dagger$.

[Dual] For any $|v\rangle \in V$, consider the linear operator $S_{|v\rangle} : \mathbb{C} \rightarrow V$ defined as follows: for any $\alpha \in \mathbb{C}$, $S_{|v\rangle}(\alpha) = \alpha |v\rangle$. Then

$$(S_{|v\rangle})^\dagger = \langle v|,$$

where $\langle v| : V \rightarrow \mathbb{C}$ is the dual linear operator corresponding to $|v\rangle$. We drop the notation $S_{|v\rangle}$ and using $|v\rangle$ to think of it as the corresponding operator and therefore write

$$|v\rangle^\dagger = \langle v|$$

.

Proof: Take any $\alpha \in \mathbb{C}$ and $|u\rangle \in V$.

$$\begin{aligned} (S_{|v\rangle}(\alpha), |u\rangle) &= (\alpha |v\rangle, |u\rangle) \\ &= (\alpha, \langle v|u\rangle) \\ &= (\alpha, \langle v|(|u\rangle)) \end{aligned}$$

Therefore, $(S_{|v\rangle})^\dagger = \langle v|$, i.e., $|v\rangle^\dagger = \langle v|$.

[Exc] For any linear operator A and $|v\rangle \in V$, we have $(A|v\rangle)^\dagger = \langle v|A^\dagger$

[Fact] For any $|v\rangle, |w\rangle \in V$, $(|w\rangle\langle v|)^\dagger = |v\rangle\langle w|$

Proof: Take any $|v_1\rangle \in V$ and $|w_1\rangle \in W$. Then

$$\begin{aligned} (|w\rangle\langle v|(|v_1\rangle), |w_1\rangle) &= (|w\rangle\langle v|v_1\rangle, |w_1\rangle) \\ &= (\langle v|v_1\rangle, |w\rangle^\dagger |w_1\rangle) \\ &= (\langle v|(|v_1\rangle), |w\rangle^\dagger |w_1\rangle) \\ &= (|v\rangle^\dagger(|v_1\rangle), |w\rangle^\dagger |w_1\rangle) \\ &= (|v_1\rangle, (|v\rangle^\dagger)^\dagger |w\rangle^\dagger |w_1\rangle) \\ &= (|v_1\rangle, |v\rangle\langle w|(|w_1\rangle)) \end{aligned}$$

Therefore, $(|w\rangle\langle v|)^\dagger = |v\rangle\langle w|$.

[Exc] Show that $(\sum_i \alpha_i A_i)^\dagger = \sum_i \alpha_i^* A_i^\dagger$.

[Self-adjoint or Hermitian] A linear operator $A : V \rightarrow V$ is self-adjoint or Hermitian if

$$A^\dagger = A$$

[Exc] The outer product $|v\rangle\langle v|$ is self-adjoint.

[Projector] Let V be an inner product space of dimension m and W be a subspace of V of dimension n . Let $\{|1\rangle, |2\rangle, \dots, |n\rangle\}$ be an orthonormal basis of W and $\{|1\rangle, \dots, |n\rangle, |n+1\rangle, \dots, |m\rangle\}$ be an orthonormal basis of V . Let $P = \sum_{i=1}^n |i\rangle\langle i|$. Then P is an orthogonal projector on the subspace W .

[Unitary Operator] An operator A is called unitary if $A^\dagger A = I$.

[Exc] If A is a unitary operator, then $AA^\dagger = I$

[Exc] Unitary operators preserve inner products, i.e., if U is a unitary operator, then, for all $|v\rangle, |w\rangle$, we have

$$(|v\rangle, |w\rangle) = (U|v\rangle, U|w\rangle)$$

Indeed, $(U|v\rangle, U|w\rangle) = (U^\dagger U|v\rangle, |w\rangle) = (I|v\rangle, |w\rangle) = (|v\rangle, |w\rangle)$.

[Outer product representation of any unitary U] Let $\{|v_i\rangle\}$ be any orthonormal basis set. Define

$|w_i\rangle = U |v_i\rangle$, so $\{|w_i\rangle\}$ is also an orthonormal basis set. Then

$$\begin{aligned}
U &= \sum_{i,j} \langle w_j | U | v_i \rangle | w_j \rangle \langle v_i | \\
&= \sum_{i,j} (| w_j \rangle, U | v_i \rangle) | w_j \rangle \langle v_i | \\
&= \sum_{i,j} (U | v_j \rangle, U | v_i \rangle) | w_j \rangle \langle v_i | \\
&= \sum_{i,j} (U^\dagger U | v_j \rangle, | v_i \rangle) | w_j \rangle \langle v_i | \\
&= \sum_{i,j} (| v_j \rangle, | v_i \rangle) | w_j \rangle \langle v_i | \\
&= \sum_i | w_i \rangle \langle v_i |
\end{aligned}$$

[Orthogonal Decomposition] A orthogonal decomposition for an operator $A : V \rightarrow V$ is a representation

$$A = \sum_i \lambda_i |i\rangle \langle i|,$$

where $\{|i\rangle\}$ form an orthonormal set of eigenvectors for A , with corresponding eigenvalues λ_i .

3 Lecture#3 (Jan 27)

The Postulates of Quantum Mechanics

Quantum mechanics provides a mathematical and conceptual framework for the development of laws a physical system must obey. In the following we give a description of the basic postulates of quantum mechanics.

Postulate 1(State Space)

- Associated to any isolated physical system is an inner product space V over $\mathbb{C}$ known as the *state space* of the system. The system is completely described by its *state vector* $|\psi\rangle \in V$, which is a unit vector, i.e., $\langle \psi | \psi \rangle = 1$.

Postulate 2 (Evolution)

- The evolution of a closed quantum system is described by a *unitary operator*. Specifically, if the system is in the state $|\psi\rangle$ at time t_1 , then its state $|\psi'\rangle$ at time t_2 is given by

$$|\psi'\rangle = U_{t_1, t_2} |\psi\rangle,$$

where U_{t_1, t_2} is a unitary operator that depends only on the times t_1 and t_2 .

Postulate 3 (Quantum Measurement)

- A quantum measurement is described by a collection of measurement operators $\{M_m\}_{m=1}^n$ acting on the state space of the system. These operators satisfy the *completeness relation*

$$\sum_{m=1}^n M_m^\dagger M_m = I$$

If the system is in the state $|\psi\rangle$ prior to measurement, then the measurement produces a classical outcome m with probability

$$\mathbb{P}[m] = \langle \psi | M_m^\dagger M_m | \psi \rangle.$$

Conditioned on obtaining classical outcome m , the post-measurement quantum state of the system is

$$|\psi_m\rangle = \frac{M_m |\psi\rangle}{\sqrt{\langle \psi | M_m^\dagger M_m | \psi \rangle}}.$$

Thus, the measurement can be viewed as a random experiment whose outcome is the pair

$$(m, |\psi_m\rangle),$$

where m represents the classical measurement outcome and $|\psi_m\rangle$ is the corresponding post-measurement quantum state.

[Exc] Show that $\sum_{m=1}^n \mathbb{P}[m] = a$

Proof:

$$\begin{aligned} 1 &= (|\psi\rangle, |\psi\rangle) \\ &= (|\psi\rangle, I |\psi\rangle) \\ &= (|\psi\rangle, (\sum_m M_m^\dagger M_m) (|\psi\rangle)) \\ &= \sum_m (|\psi\rangle, M_m^\dagger M_m (|\psi\rangle)) \\ &= \sum_m \langle \psi | M_m^\dagger M_m | \psi \rangle \\ &= \sum_m \mathbb{P}[m] \end{aligned}$$

4 Lecture#4 (Jan 30)

[Exc] Consider a quantum mechanical system with state space $\mathbb{C}^2$ (with $\{|0\rangle, |1\rangle\}$ as basis) and the current state vector $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ with $\langle \psi | \psi \rangle = |\alpha|^2 + |\beta|^2 = 1$. Suppose we attempt to measure the state using the following collection of measurement operators given by the outer products $\{M_0 = |0\rangle\langle 0|, M_1 = |1\rangle\langle 1|\}$ satisfying completeness property. As per the Postulate #3, the possible set of post-measurement outputs are:

$(0, |\psi_0\rangle = \frac{M_0 \psi}{\sqrt{\langle \psi | M_0^\dagger M_0 | \psi \rangle}})$ with probability $\langle \psi | M_0^\dagger M_0 | \psi \rangle$, and $(1, |\psi_1\rangle = \frac{M_1 \psi}{\sqrt{\langle \psi | M_1^\dagger M_1 | \psi \rangle}})$ with probability $\langle \psi | M_1^\dagger M_1 | \psi \rangle$.

Check that

$$\begin{aligned} \langle \psi | M_0^\dagger M_0 | \psi \rangle &= (|\psi\rangle, M_0^\dagger M_0 |\psi\rangle) \\ &= (M_0 |\psi\rangle, M_0 |\psi\rangle) \\ &= ((|0\rangle\langle 0|)(\alpha |0\rangle + \beta |1\rangle), (|0\rangle\langle 0|)(\alpha |0\rangle + \beta |1\rangle)) \\ &= (\alpha(|0\rangle\langle 0|)(|0\rangle) + \beta(|0\rangle\langle 0|)(|1\rangle), \alpha(|0\rangle\langle 0|)(|0\rangle) + \beta(|0\rangle\langle 0|)(|1\rangle)) \\ &= (\alpha |0\rangle\langle 0|0\rangle + \beta |0\rangle\langle 0|1\rangle, \alpha |0\rangle\langle 0|0\rangle + \beta |0\rangle\langle 0|1\rangle) \\ &= (\alpha |0\rangle, \alpha |0\rangle) \\ &= \alpha^* \alpha (|0\rangle, |0\rangle) \\ &= |\alpha|^2 \end{aligned}$$

Similarly, $\langle \psi | M_1^\dagger M_1 | \psi \rangle = |\beta|^2$. Therefore, $|\psi_0\rangle = \frac{M_0 \psi}{\sqrt{\langle \psi | M_0^\dagger M_0 | \psi \rangle}} = \frac{\alpha |0\rangle}{|\alpha|}$ and $|\psi_1\rangle = \frac{M_1 \psi}{\sqrt{\langle \psi | M_1^\dagger M_1 | \psi \rangle}} = \frac{\beta |1\rangle}{|\beta|}$

Postulate 4 (Composite Systems)

Suppose a composite quantum system made up of two (or more) distinct physical systems. The state space of this composite system is the tensor product spaces of the component physical systems. Moreover, if we have systems numbered 1 through n , and system number i is prepared in the state $|\psi\rangle$, then the joint state of the total system is $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$.

Tensor Products

Let V, W be vector spaces over $\mathbb{C}$.

- For every $|v\rangle \in V$ and $|w\rangle \in W$, introduce a formal symbol $|v\rangle \otimes |w\rangle$. Let F be the set of all finite formal linear combinations

$$F = \left\{ \sum_{k=1}^r \alpha_k |v_k\rangle \otimes |w_k\rangle \mid r \in \mathbb{N}, \alpha_k \in \mathbb{C}, |v_k\rangle \in V, |w_k\rangle \in W \right\}$$

Addition and scalar multiplication are defined term wise, so F is a vector space (the *free vector space* generated by the symbols $|v\rangle \otimes |w\rangle$). Note that the zero element in F is given by $\sum_k 0(|v_k\rangle \otimes |w_k\rangle)$

- Let $R \subseteq F$ be the subspace generated by the following elements: for every $|v\rangle, |v_1\rangle, |v_2\rangle \in V$, $|w\rangle, |w_1\rangle, |w_2\rangle \in W$, and $\alpha \in \mathbb{C}$

$$\begin{aligned} & \alpha(|v\rangle \otimes |w\rangle) - (\alpha |v\rangle) \otimes |w\rangle, \\ & \alpha(|v\rangle \otimes |w\rangle) - |v\rangle \otimes (\alpha |w\rangle), \\ & (|v_1\rangle + |v_2\rangle) \otimes |w\rangle - (|v_1\rangle \otimes |w\rangle) + (|v_2\rangle \otimes |w\rangle), \\ & |v\rangle \otimes (|w_1\rangle + |w_2\rangle) - (|v\rangle \otimes |w_1\rangle) + (|v\rangle \otimes |w_2\rangle) \end{aligned}$$

Finally, the tensor product is defined as the quotient vector space

$$V \otimes W = F/R$$

[Fact] Let $\{|i\rangle\}_{i=1}^m$ and $\{|j\rangle\}_{j=1}^n$ be the bases of V and W respectively. The basis of $V \otimes W$ is given by $\{|i\rangle \otimes |j\rangle\}_{1 \leq i \leq m, 1 \leq j \leq n}$. Therefore dimension of $V \otimes W$ is mn .

Brief Sketch: Take any $|v\rangle \otimes |w\rangle$. Suppose $|v\rangle = \sum \alpha_i |i\rangle$ and $|w\rangle = \sum \beta_j |j\rangle$. Then

$$\begin{aligned} |v\rangle \otimes |w\rangle &= \left(\sum \alpha_i |i\rangle \right) \otimes \left(\sum \beta_j |j\rangle \right) \\ &= \sum_i (\alpha_i |i\rangle) \otimes \left(\sum \beta_j |j\rangle \right) \\ &= \sum_{i,j} (\alpha_i |i\rangle) \otimes (\beta_j |j\rangle) \\ &= \sum_{i,j} \alpha_i (|i\rangle \otimes (\beta_j |j\rangle)) \\ &= \sum_{i,j} \alpha_i \beta_j (|i\rangle \otimes |j\rangle) \end{aligned}$$

[Exc] If V and W are inner product spaces over $\mathbb{C}$, then $V \otimes W$ is also an inner product space, where the inner product is given by

$$\left(\sum_i \alpha_i |v_i\rangle \otimes |w_i\rangle, \sum_j \beta_j |v'_j\rangle \otimes |w'_j\rangle \right) = \sum_{i,j} \alpha_i^* \beta_j (|v_i\rangle, |v'_j\rangle) (|w_i\rangle, |w'_j\rangle)$$

[Exc] Consider the vector space $\mathbb{C}^2$ with orthonormal basis $\{|0\rangle, |1\rangle\}$. Consider the tensor product space $\mathbb{C}^2 \otimes \mathbb{C}^2$. Its basis is given by

$$\{|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle\}$$

We use the following natural labels for the basis elements:

$$\begin{aligned} |00\rangle &= |0\rangle \otimes |0\rangle \\ |01\rangle &= |0\rangle \otimes |1\rangle \\ |10\rangle &= |1\rangle \otimes |0\rangle \\ |11\rangle &= |1\rangle \otimes |1\rangle \end{aligned}$$

[Exc] Not every element of the tensor product space $V \otimes W$ can be written as a simple tensor of the form $|v\rangle \otimes |w\rangle$, with $|v\rangle \in V$ and $|w\rangle \in W$.

Consider the space $\mathbb{C}^2 \otimes \mathbb{C}^2$ and the vector $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$. We claim that this vector cannot be expressed as $|v\rangle \otimes |w\rangle$, for any $|v\rangle, |w\rangle \in \mathbb{C}^2$. Suppose, for the sake of contradiction, that $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle = |v\rangle \otimes |w\rangle$ where $|v\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$ and $|w\rangle = \beta_0|0\rangle + \beta_1|1\rangle$. Then

$$\begin{aligned} \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle &= (\alpha_0|0\rangle + \alpha_1|1\rangle) \otimes (\beta_0|0\rangle + \beta_1|1\rangle) \\ &= \alpha_0\beta_0|0\rangle \otimes |0\rangle + \alpha_0\beta_1|0\rangle \otimes |1\rangle + \alpha_1\beta_0|1\rangle \otimes |0\rangle + \alpha_1\beta_1|1\rangle \otimes |1\rangle \\ &= \alpha_0\beta_0|00\rangle + \alpha_0\beta_1|01\rangle + \alpha_1\beta_0|10\rangle + \alpha_1\beta_1|11\rangle \end{aligned}$$

This implies, $\alpha_0\beta_0 = 1$, $\alpha_0\beta_1 = 0$, $\alpha_1\beta_0 = 0$, and $\alpha_1\beta_1 = 1$. These equations cannot be satisfied simultaneously. Hence, $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$ can not be written as a simple tensor $|v\rangle \otimes |w\rangle$.

[Tensor of Linear Operators]

Let V and W be two vector space with bases $\{|i\rangle\}$ and $\{|j\rangle\}$ respectively. Take any linear operators $A : V \rightarrow V'$ and $B : W \rightarrow W'$. The tensor product of A and B , denoted as $A \otimes B$, is a linear map defined as follows: for any basis element $|i\rangle \otimes |j\rangle$, define $(A \otimes B)(|i\rangle \otimes |j\rangle) = A|i\rangle \otimes B|j\rangle$, and extend linearly, i.e, for any $\sum \alpha_{i,j} |i\rangle \otimes |j\rangle$, $(A \otimes B)(\sum \alpha_{i,j} |i\rangle \otimes |j\rangle) = \sum \alpha_{i,j} (A \otimes B)(|i\rangle \otimes |j\rangle)$.

[Exc] Show that $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$.

Take any $|v\rangle \in V, |w\rangle \in W, |v'\rangle \in V', |w'\rangle \in W'$, we have

$$\begin{aligned} ((A \otimes B)(|v\rangle \otimes |w\rangle), |v'\rangle \otimes |w'\rangle) &= (A|v\rangle \otimes B|w\rangle, |v'\rangle \otimes |w'\rangle) \\ &= (A|v\rangle, |v'\rangle) (B|w\rangle, |w'\rangle) \\ &= (|v\rangle, A^\dagger |v'\rangle) (|w\rangle, B^\dagger |w'\rangle) \\ &= (|v\rangle \otimes |w\rangle, A^\dagger |v'\rangle \otimes B^\dagger |w'\rangle) \\ &= (|v\rangle \otimes |w\rangle, (A^\dagger \otimes B^\dagger)(|v'\rangle \otimes |w'\rangle)) \end{aligned}$$

Therefore, $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$.

5 Lecture#5 (Feb 3)

[Single Qubit Gate] A single qubit gate is a linear operator on $\mathbb{C}^2$.

In the following we discuss a few well-known single qubit gates

[Quantum NOT Gate] The quantum NOT gate, denoted by X , is given by the linear operator whose matrix representation, with respect to the standard basis $\{|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}\}$, is

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

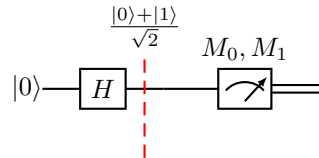
Input	Gate Action	Output
$ 0\rangle$	$\xrightarrow{X}$	$ 1\rangle$
$ 1\rangle$	$\xrightarrow{X}$	$ 0\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$\xrightarrow{X}$	$\beta 0\rangle + \alpha 1\rangle$

[Hadamard Gate] The Hadamard gate, denoted by H , is given by the linear operator whose matrix representation, with respect to the standard basis $\{|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}\}$, is

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Input	Gate Action	Output
$ 0\rangle$	$\xrightarrow{H}$	$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$
$ 1\rangle$	$\xrightarrow{H}$	$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$
$\alpha 0\rangle + \beta 1\rangle$	$\xrightarrow{H}$	$\frac{\alpha + \beta}{\sqrt{2}} 0\rangle + \frac{\alpha - \beta}{\sqrt{2}} 1\rangle$

[Quantum Random Bit Generator]

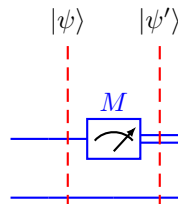


[Two Qubit Gate] A two qubit gate is a linear operator on the tensor space $\mathbb{C}^2 \otimes \mathbb{C}^2$.

6 Lecture# (Feb 17)

6.1 Partial Measurement

In many quantum protocols, we measure only a part of a composite system while leaving the remaining subsystem untouched. Such a measurement is called a partial measurement. Formally, this is implemented by tensoring the measurement operators acting on one subsystem with the identity operator on the other subsystem. We now illustrate this for a two-qubit system.



Let $|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$ be a general two-qubit state. Let $\{M_0, M_1\}$ be the measurement operators acting on a single qubit. Consider the operators $\{M_0 \otimes I, M_1 \otimes I\}$. Check that $\{M_0 \otimes I, M_1 \otimes I\}$ satisfies the completeness property.

A measurement described by $\{M_0 \otimes I, M_1 \otimes I\}$ is therefore a partial measurement on the first-qubit. The possible outcomes are 0 or 1. The probability of 0-outcome is

$$\begin{aligned} ((M_0 \otimes I)|\psi\rangle, (M_0 \otimes I)|\psi\rangle) &= (\alpha_{00}|00\rangle + \alpha_{01}|01\rangle, \alpha_{00}|00\rangle + \alpha_{01}|01\rangle) \\ &= |\alpha_{00}|^2 + |\alpha_{01}|^2 \end{aligned}$$

Conditioned on 0-outcome, the post-measurement (collapsed) state is

$$\begin{aligned} \frac{(M_0 \otimes I)|\psi\rangle}{\sqrt{((M_0 \otimes I)|\psi\rangle, (M_0 \otimes I)|\psi\rangle)}} &= \frac{(M_0 \otimes I)|\psi\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}} \\ &= \frac{\alpha_{00}|00\rangle + \alpha_{01}|01\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}} \end{aligned}$$

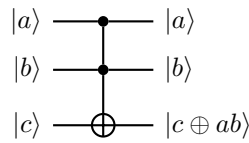
Similarly, the probability of obtaining outcome 1 is $|\alpha_{10}|^2 + |\alpha_{11}|^2$, and conditioned on this outcome the collapsed state is

$$\frac{\alpha_{10}|10\rangle + \alpha_{11}|11\rangle}{\sqrt{|\alpha_{10}|^2 + |\alpha_{11}|^2}}$$

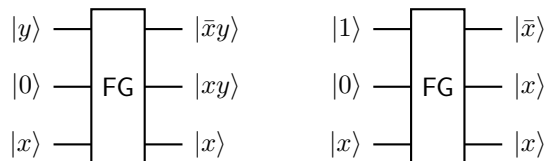
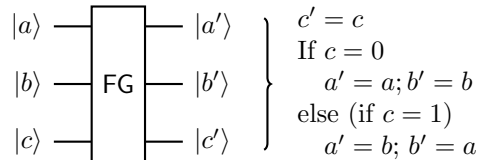
7 Lecture# (Feb 19)

The Toffoli and Fredkin gates are important three-qubit gates in quantum computing. Both gates are reversible and perform controlled operations based on the values of control qubits. The Toffoli gate (also called the controlled-controlled-NOT gate) flips a target qubit only when both control qubits are in the state $|1\rangle$, while the Fredkin gate (controlled-SWAP gate) swaps two qubits conditioned on the value of a control qubit.

[Toffoli Gate]

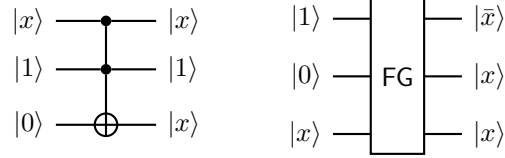


[Fredkin Gate]

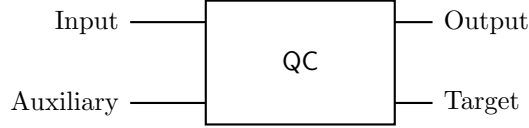


7.1 Characterisation of Cloanable States

The following quantum circuits (Toffoli and Fredkin gates) takes the input state $|x\rangle$, where $x \in \{0, 1\}$, and copy it onto two output wires. Thus, these gates appear to clone the input state. A natural question is whether these circuits can also clone a superposition state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. The answer is no: both circuits only clone the computational basis states $|0\rangle$ and $|1\rangle$.

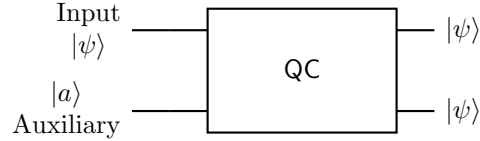


We now characterise the set of states that any quantum circuit could possibly clone. Without loss of generality, we assume the circuit has the following structure.



Let the entire quantum circuit QC be represented by a unitary operator U . The second register is an auxiliary system prepared in a fixed state $|a\rangle$, which may itself consist of one or more qubits (that is, the auxiliary input can be a multi-qubit state), and is assumed to be normalized, i.e., $\langle a|a\rangle = 1$.

Suppose $|\psi\rangle$ is a state that is cloned by QC, that is



Thus, $U(|\psi\rangle \otimes |a\rangle) = |\psi\rangle \otimes |\psi\rangle$.

Now suppose $|\phi\rangle$ is another state that is also cloned by the same circuit QC. Then, $U(|\phi\rangle \otimes |a\rangle) = |\phi\rangle \otimes |\phi\rangle$.

Since U is unitary, it preserves inner products. Therefore,

$$\begin{aligned} (U(|\psi\rangle \otimes |a\rangle), U(|\phi\rangle \otimes |a\rangle)) &= (|\psi\rangle \otimes |a\rangle, |\phi\rangle \otimes |a\rangle) \\ &= \langle \psi|\phi\rangle \langle a|a\rangle \\ &= \langle \psi|\phi\rangle \quad // * \quad \langle a|a\rangle = 1 \end{aligned}$$

On the other hand,

$$\begin{aligned} (|\psi\rangle \otimes |\psi\rangle, |\phi\rangle \otimes |\phi\rangle) &= \langle \psi|\phi\rangle \langle \psi|\phi\rangle \\ &= (\langle \psi|\phi\rangle)^2 \end{aligned}$$

This implies, $(\langle \psi|\phi\rangle)^2 = \langle \psi|\phi\rangle$. Which means, either $\langle \psi|\phi\rangle = 0$ or $\langle \psi|\phi\rangle = 1$. The former implies $|\phi\rangle$ is orthogonal to $|\psi\rangle$, while the latter implies $|\phi\rangle = |\psi\rangle$.

Therefore, the set of states that can be cloned by a unitary quantum circuit must consist of mutually orthogonal states.

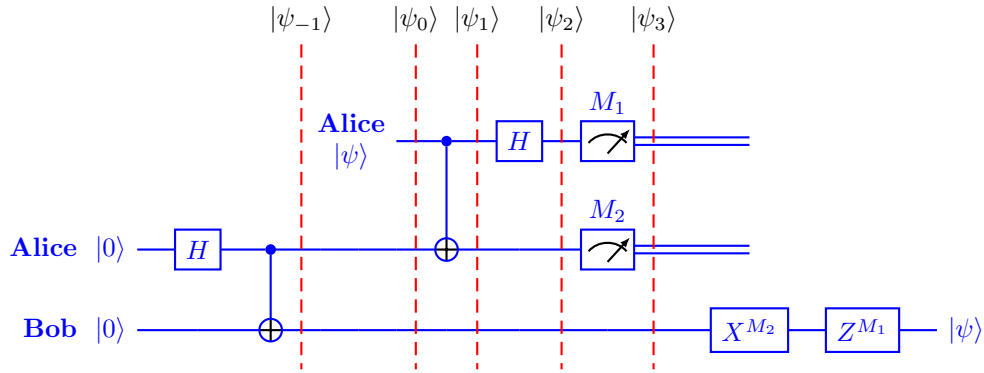
8 Quantum Teleportation

Suppose Alice possesses an unknown qubit state

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

and wishes to transmit this state to Bob, who is spatially separated from her. Since the state is unknown, it cannot be copied or directly measured without disturbance. The task is to transfer the quantum state $|\psi\rangle$ to Bob using prior shared entanglement and classical communication.

While they were together, Alice and Bob generated an EPR pair and each kept one qubit of the entangled pair before separating. Later, Alice is given an unknown qubit state $|\psi\rangle$ and her task is to transmit this state to Bob without physically sending the qubit itself.



[State $|\psi_{-1}\rangle$] Clearly,

$$|\psi_{-1}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

[State $|\psi_0\rangle$] Suppose $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$. Then

$$\begin{aligned} |\psi_0\rangle &= |\psi\rangle \otimes \left(\frac{|00\rangle + |11\rangle}{\sqrt{2}} \right) \\ &= (\alpha |0\rangle + \beta |1\rangle) \left(\frac{|00\rangle + |11\rangle}{\sqrt{2}} \right) \\ &= (\alpha |0\rangle + \beta |1\rangle) \left(\frac{|0\rangle |0\rangle + |1\rangle |1\rangle}{\sqrt{2}} \right) \\ &= \frac{1}{\sqrt{2}} [(\alpha |00\rangle + \beta |10\rangle) |0\rangle + (\alpha |01\rangle + \beta |11\rangle) |1\rangle] \end{aligned}$$

[State $|\psi_1\rangle$]

$$\begin{aligned} |\psi_1\rangle &= (\text{C-NOT} \otimes I)(|\psi_0\rangle) \\ &= (\text{C-NOT} \otimes I) \left(\frac{1}{\sqrt{2}} [(\alpha |00\rangle + \beta |10\rangle) |0\rangle + (\alpha |01\rangle + \beta |11\rangle) |1\rangle] \right) \\ &= \frac{1}{\sqrt{2}} [(\alpha |00\rangle + \beta |11\rangle) |0\rangle + (\alpha |01\rangle + \beta |10\rangle) |1\rangle] \\ &= \frac{1}{\sqrt{2}} [(\alpha |000\rangle + \beta |110\rangle) + (\alpha |011\rangle + \beta |101\rangle)] \\ &= \frac{1}{\sqrt{2}} [(\alpha |0\rangle |00\rangle + \beta |1\rangle |10\rangle) + (\alpha |0\rangle |11\rangle + \beta |1\rangle |01\rangle)] \\ &= \frac{1}{\sqrt{2}} [\alpha |0\rangle (|00\rangle + |11\rangle) + \beta |1\rangle (|10\rangle + |01\rangle)] \end{aligned}$$

[State $|\psi_2\rangle$]

$$\begin{aligned}
|\psi_2\rangle &= (H \otimes I^2)(|\psi_1\rangle) \\
&= (H \otimes I^2)\left(\frac{1}{\sqrt{2}}[\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|10\rangle + |01\rangle)]\right) \\
&= \frac{1}{\sqrt{2}}\left[\alpha\frac{|0\rangle + |1\rangle}{\sqrt{2}}(|00\rangle + |11\rangle) + \beta\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)(|10\rangle + |01\rangle)\right] \\
&= \frac{1}{2}[\alpha(|000\rangle + |011\rangle + |100\rangle + |111\rangle) + \beta(|010\rangle + |001\rangle - |110\rangle - |101\rangle)] \\
&= \frac{1}{2}[\alpha(|00\rangle|0\rangle + |01\rangle|1\rangle + |10\rangle|0\rangle + |11\rangle|1\rangle) + \beta(|01\rangle|0\rangle + |00\rangle|1\rangle - |11\rangle|0\rangle - |10\rangle|1\rangle)] \\
&= \frac{1}{2}[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)]
\end{aligned}$$

[State $|\psi_3\rangle$ (After Alice's measurement on its first two qubits)]

$$|\psi_3\rangle = (M_1 \otimes M_2 \otimes I)(|\psi_2\rangle)$$

where possible outcomes of M_1, M_2 are $\{0, 1\}$ and therefore possible outcome of $M_1 \otimes M_2$ are $\{00, 01, 10, 11\}$.

Case 1: Suppose the outcome of Alice's measurement $M_1 \otimes M_2$ is 00. Then, by the postulates of partial measurement, the collapsed state becomes

$$\frac{|00\rangle(\alpha|0\rangle + \beta|1\rangle)}{\sqrt{(1/2)^2}} = |00\rangle(\alpha|0\rangle + \beta|1\rangle) = |00\rangle|\psi\rangle$$

If Bob is informed that Alice's measurement outcome is 00, he immediately knows that his qubit is already in the state $|\psi\rangle$ - the state that Alice intended to transmit.

Case 2: Suppose the outcome of Alice's measurement $M_1 \otimes M_2$ is 01. Then the collapsed state is

$$|01\rangle(\alpha|1\rangle + \beta|0\rangle)$$

If Bob is informed that the measurement outcome is 01, he can recover $|\psi\rangle$ by applying the X -gate to his qubit. Indeed, $X(\alpha|1\rangle + \beta|0\rangle) = \alpha|0\rangle + \beta|1\rangle = \psi$.

Case 3: Suppose the outcome of Alice's measurement $M_1 \otimes M_2$ is 10. Then the collapsed state is

$$|10\rangle(\alpha|0\rangle - \beta|1\rangle)$$

If Bob is informed that the measurement outcome is 10, he can recover $|\psi\rangle$ by applying $Z(\alpha|0\rangle - \beta|1\rangle) = \alpha|0\rangle + \beta|1\rangle = \psi$.

Case 4: Suppose the outcome of Alice's measurement $M_1 \otimes M_2$ is 11. Then the collapsed state is

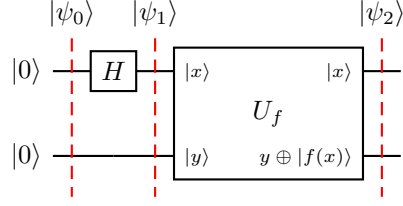
$$|11\rangle(\alpha|1\rangle - \beta|0\rangle)$$

If Bob is informed that the measurement outcome is 11, he can recover $|\psi\rangle$ by applying $ZX(\alpha|1\rangle - \beta|0\rangle) = Z(X(\alpha|1\rangle - \beta|0\rangle)) = Z(\alpha|0\rangle - \beta|1\rangle) = \alpha|0\rangle + \beta|1\rangle = \psi$.

9 Quantum Parallelism

One of the remarkable features of quantum computation is quantum parallelism, which allows a quantum circuit to evaluate a function on multiple inputs simultaneously. By preparing a superposition of inputs

and applying a unitary transformation that computes the function, the quantum system processes all inputs at once. The following circuit illustrates this idea for a simple function ($f : \{0, 1\} \rightarrow \{0, 1\}$) evaluation.



$$|\psi_0\rangle = |0\rangle |0\rangle$$

$$\begin{aligned} |\psi_1\rangle &= H(|0\rangle) |0\rangle \\ &= \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle \\ &= \frac{|00\rangle + |10\rangle}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} |\psi_2\rangle &= U_f(|\psi_1\rangle) \\ &= U_f\left(\frac{|00\rangle + |10\rangle}{\sqrt{2}}\right) \\ &= \frac{|0f(0)\rangle + |0f(1)\rangle}{\sqrt{2}} \end{aligned}$$

9.1 The Deutsch Algorithm

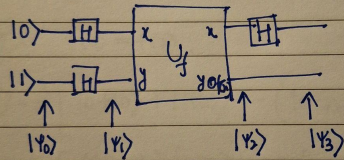
Deutsch's Algorithm

Given $f: \{0,1\} \rightarrow \{0,1\}$.

Let U_f be a quantum circuit defined by

$$U_f(|x\rangle|y\rangle) = |x\rangle|y \oplus f(x)\rangle$$

In the following we discuss a quantum circuit computing $f(0) \oplus f(1)$, using only one evaluation of f .



$$|\psi_0\rangle = |0\rangle$$

$$\begin{aligned} |\psi_1\rangle &= H^{\otimes 2}(|01\rangle) = H(|0\rangle)H(|1\rangle) \\ &= \left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right) \\ &= \frac{|00\rangle - |01\rangle + |10\rangle - |11\rangle}{2} \end{aligned}$$

$$|\psi_2\rangle = U_f(|\psi_1\rangle)$$

$$= U_f\left(\frac{|00\rangle - |01\rangle + |10\rangle - |11\rangle}{2}\right)$$

$$= \frac{1}{2} \left[U_f(|00\rangle) - U_f(|01\rangle) + U_f(|10\rangle) - U_f(|11\rangle) \right]$$

$$= \frac{1}{2} \left[|0f(0)\rangle - |0\bar{f}(0)\rangle + |1f(1)\rangle - |1\bar{f}(1)\rangle \right]$$

Case 1: $f(0) = f(1)$.

$$|\psi_2\rangle = \frac{1}{2} \left[|0f(0)\rangle + |1f(1)\rangle - (|0\bar{f}(0)\rangle + |1\bar{f}(1)\rangle) \right]$$

$$= \frac{1}{2} \left[(|0\rangle+|1\rangle)(|f(0)\rangle) - (|0\rangle+|1\rangle)(|f(0)\rangle) \right]$$

$$= \frac{1}{2} \left[(|0\rangle+|1\rangle)(|f(0)\rangle - |f(0)\rangle) \right]$$

$$= \pm \frac{1}{2} \left[(|0\rangle+|1\rangle)(|0\rangle-|1\rangle) \right]$$

$$= \pm \left(\frac{|0\rangle+|1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle-|1\rangle}{\sqrt{2}} \right)$$

Case 2: $f(0) \neq f(1)$.

Show that

$$|\psi_2\rangle = \pm \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Therefore,

$$|\psi_2\rangle = \begin{cases} \pm \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) & \text{if } f(0) = f(1) \\ \pm \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) & \text{if } f(0) \neq f(1) \end{cases}$$

Finally, $|\psi_3\rangle = H \otimes I (|\psi_2\rangle)$

Case A: $f(0) = f(1)$

$$\begin{aligned} |\psi_3\rangle &= \pm \left[\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) + \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right] \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= \pm |0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \end{aligned}$$

Case B: $f(0) \neq f(1)$

Show that

$$|\psi_3\rangle = \pm |1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Check that

$$\begin{aligned} f(0) \oplus f(1) &= 0 & \text{if } f(0) = f(1) \\ &= 1 & \text{if } f(0) \neq f(1) \end{aligned}$$

Therefore

$$|\psi_3\rangle = \pm |f(0) \oplus f(1)\rangle \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right]$$

—xx—xx—

9.2 The Deutsch-Jozsa Algorithm

①

The Deutsch-Jozsa Algorithm

The Deutsch's Problem.

Bob

Input: $f: \{0,1\}^n \rightarrow \{0,1\}$
 $\tau \in \{0,1\}$

Where
 $\tau = \begin{cases} 1 & \text{if } f \text{ is constant} \\ 0 & \text{if } f \text{ is balanced} \end{cases}$

Alice

Bob Computer

$x_i (x_i \in \{0,1\}^n)$ for $i=1$ to q

$y_i = f(x_i)$

$\tau' \leftarrow \text{Alice}(\{(x_i, y_i)\}_{i=1}^q)$

Alice wins if
 $\tau' = \tau$

②

Let U_f be a unitary operator st.
for any $x \in \{0,1\}^n, y \in \{0,1\}$

$$U_f(|x\rangle \otimes |y\rangle) = |x\rangle \otimes |y \oplus f(x)\rangle$$

We Now describe the Deutsch-Jozsa algorithm to solve the above problem.

Step 1: Alice Prepares the following Composite System.

Quantum circuit diagram showing two registers. The top register starts with $|0\rangle^{\otimes n}$ and the bottom register starts with $|1\rangle$. Both registers pass through Hadamard gates. The top register then passes through a unitary operator U_f . The bottom register is initialized to $|1\rangle$ and the top register is initialized to $|0\rangle^{\otimes n}$.

Note: $|0\rangle^{\otimes n} = |0\rangle \otimes \dots \otimes |0\rangle$

$\text{---}^n \equiv$ denotes n -quantum wires

$H^{\otimes n} = \underbrace{H \otimes \dots \otimes H}_n$

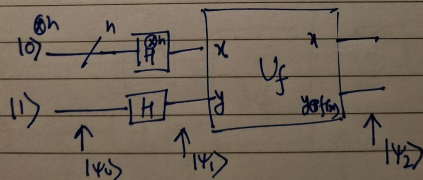
Lets Compute $|\psi_0\rangle$ and $|\psi_1\rangle$.

$|\psi_0\rangle = |0\rangle^{\otimes n} \otimes |1\rangle$

③

$$\begin{aligned}
 |\psi_1\rangle &= H^{\otimes n} \otimes H \left(|\psi_0\rangle \right) \\
 &= (H^{\otimes n} \otimes H) \left(|0\rangle^{\otimes n} \otimes |1\rangle \right) \\
 &= H^{\otimes n} \left(|0\rangle^{\otimes n} \right) \otimes H(|1\rangle) \\
 &= \left(\sum_{x \in \{0,1\}^n} \frac{|x\rangle}{2^{n/2}} \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\
 &= \sum_{x \in \{0,1\}^n} \frac{|x\rangle}{2^{n/2}} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)
 \end{aligned}$$

Step 2: Alice let Bob evaluate $f(x)$ using U_f and quantum parallelism on the given circuit as follows.

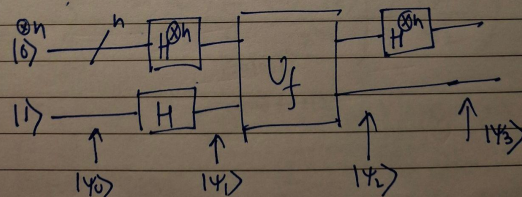


Lets compute $|\psi_2\rangle$

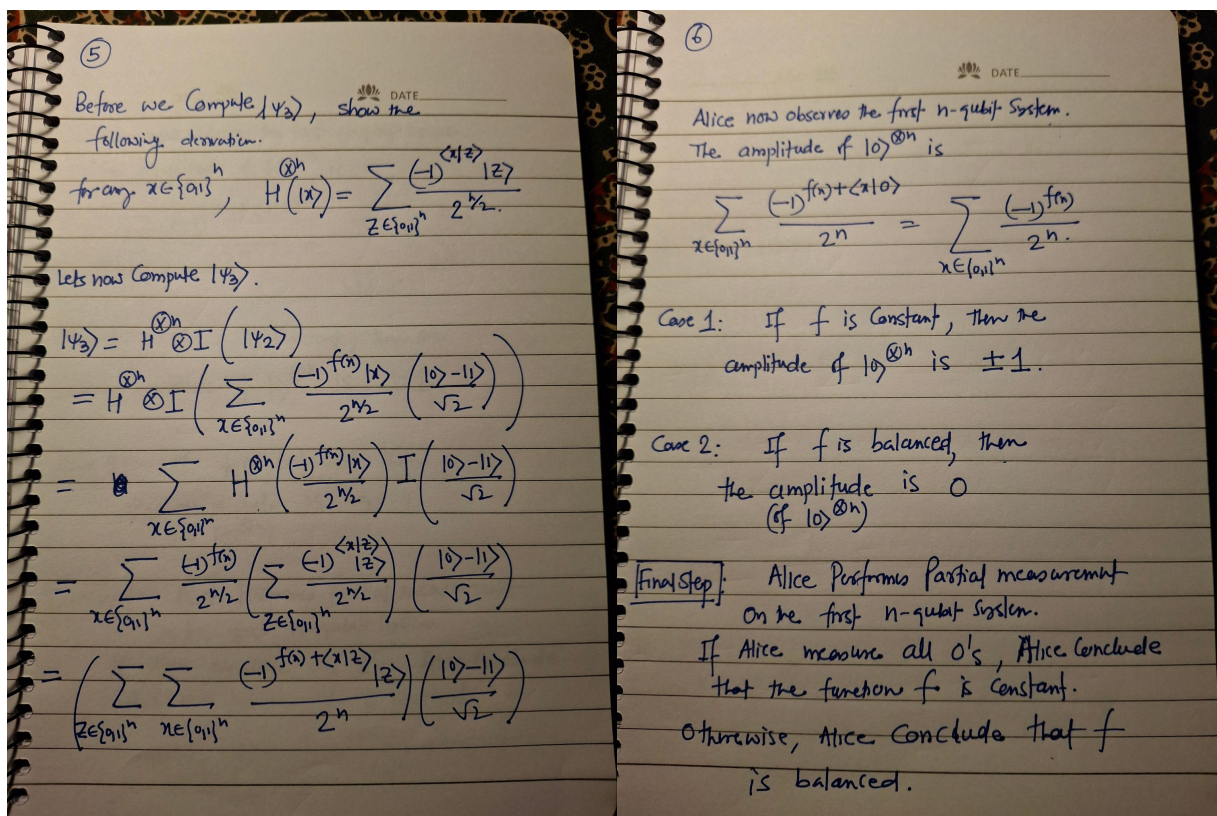
④

$$\begin{aligned}
 |\psi_2\rangle &= U_f \left(|\psi_1\rangle \right) \\
 &= U_f \left(\sum_{x \in \{0,1\}^n} \frac{|x\rangle}{2^{n/2}} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right) \\
 &= \sum_{x \in \{0,1\}^n} \frac{|x\rangle}{2^{n/2}} \left(\frac{|f(x)\rangle - |f(x)\rangle}{\sqrt{2}} \right) \\
 &= \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \frac{|x\rangle}{2^{n/2}} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)
 \end{aligned}$$

Step 3: Alice now takes over the system and evaluate it as follows:



Lets compute $|\psi_3\rangle$.



10 Projective Measurements

We now discuss an important special case of the general measurement postulate, **Postulate 3**.

Projective Measurements:

A projective measurement is described by an operator M (called *observable*), a Hermitian operator on the state space of the system being observed. Suppose the M has a spectral decomposition,

$$M = \sum_m \lambda_m P_m,$$

where P_m is the projector onto the eigenspace of M with eigenvalue λ_m . The possible outcomes of the measurement correspond to the eigenvalues, λ_m , of M . Upon measuring the state $|\psi\rangle$, the probability of getting result λ_m is given by

$$\mathbb{P}[\lambda_m] = \langle \psi | P_m | \psi \rangle$$

Given that outcome λ_m occurred, the state of the quantum system immediately after the measurement is

$$\frac{P_m |\psi\rangle}{\sqrt{\mathbb{P}[\lambda_m]}}$$

[Projective Measurement as a special case of Postulate 3] Suppose the measurement operators $\{M_m\}$ in Postulate 3, in addition to satisfying the completeness relation $\sum_m M_m^\dagger M_m = I$, also satisfy the conditions that M_m are orthogonal projectors, that is, the M_m are Hermitian, $M_m^2 = M_m$, and $M_i M_j = 0$

when $i \neq j$. With these additional restrictions, Postulate 3 reduces to a projective measurement as just defined.

[Examples of Projective Measurement on Single Qubit System]

[Example] Consider the observable $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Check that Z has eigenvalues $+1$ and -1 with corresponding eigenvectors $|0\rangle$ and $|1\rangle$ respectively. Therefore, $Z = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|$. Thus, for example, measurement of Z on the state $|\psi\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ gives result $+1$ with $\mathbb{P}[+1] = \langle\psi|(|0\rangle\langle 0|)|\psi\rangle = (\langle\psi|, |0\rangle\langle 0|(|\psi\rangle)) = (\langle\psi|, |0\rangle\langle 0|\psi\rangle) = (\langle\psi|, |0\rangle\langle 0| \frac{|0\rangle+|1\rangle}{\sqrt{2}}) = (\langle\psi|, |0\rangle\langle 0| \frac{1+0}{\sqrt{2}}) = (\langle\psi|, |0\rangle\langle 0| \frac{1}{\sqrt{2}}) = (\langle\psi|, \frac{|0\rangle}{\sqrt{2}}) = (\frac{|0\rangle+|1\rangle}{\sqrt{2}}, \frac{|0\rangle}{\sqrt{2}}) = \frac{1}{2}$. Similarly, $\mathbb{P}[-1] = \frac{1}{2}$.

[Example] For any $\hat{v} \in \mathbb{R}^3$ (with $||\hat{v}|| = 1$), consider the following observable:

$$\hat{v} \cdot \hat{\sigma} = v_1\sigma_1 + v_2\sigma_2 + v_3\sigma_3 = v_1X + v_2Y + v_3Z$$

where $\sigma_1 = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\sigma_2 = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, and $\sigma_3 = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

Measurement of this observable $\hat{v} \cdot \hat{\sigma}$ is sometimes referred to as a *measurement of spin along the $\hat{v}$ axis*.

[Exc] If A be a linear operator on V such that $A^2 = I$. Show that A has eigenvalues ± 1 , and that the projectors onto the corresponding eigenspaces are given by $A_{\pm} = \frac{I \pm A}{2}$.

Proof. Let v be an eigenvector with eigenvalue λ_v . This means $Av = \lambda_v v$. Which mean $A^2v = AA v = A\lambda_v v = \lambda_v Av = \lambda_v \lambda_v v = \lambda_v^2 v$. But $A^2 = I$, which means $\lambda_v^2 v = v$. Therefore, $\lambda_v^2 = 1$, implying $\lambda_v = \pm 1$.

Define $V_+ = \text{Ker}(A - I)$ and $V_- = \text{Ker}(A + I)$. Check that V is a direct sum of the subspaces V_+ and V_- , i.e., for every $w \in V$, there exists uniques $v \in V_+$ and $u \in V_-$ such that $w = v + u$.

Define $A_+ = \frac{I+A}{2}$ and $A_- = \frac{I-A}{2}$. We now show that A_+ is the projector on the eigenspace of the eigenvalue $+1$ (similarly, A_- is the projector on the eigenspace of the eigenvalue -1). Take any eigenvector u from the eigenspace of $+1$, i.e., $u \in V_+$.

$$\begin{aligned} A_+(u) &= \frac{I+A}{2}(u) \\ &= \frac{I(u) + A(u)}{2} \\ &= \frac{u + (+1)u}{2} \\ &= \frac{2u}{2} = u \end{aligned}$$

Take any $v \in V_-$. Then

$$\begin{aligned} A_+(v) &= \frac{I+A}{2}(v) \\ &= \frac{I(v) + A(v)}{2} \\ &= \frac{v - v}{2} \\ &= 0 \end{aligned}$$

[Exc] Show that, for any $\hat{v} \in \mathbb{R}^3$, the operator $\hat{v} \cdot \hat{\sigma}$ satisfies $(\hat{v} \cdot \hat{\sigma})^2 = I$. Therefore, $\hat{v} \cdot \hat{\sigma}$ has eigenvalues ± 1 and its diagonal representation is given by

$$\hat{v} \cdot \hat{\sigma} = \frac{I + \hat{v} \cdot \hat{\sigma}}{2} - \frac{I - \hat{v} \cdot \hat{\sigma}}{2}$$

[Exc] Suppose the single qubit system's current state is $|0\rangle$. Consider the observable $\hat{v} \cdot \hat{\sigma}$ for a measurement. Compute $\mathbb{P}[+1]$, the probability of obtaining the result $+1$. What is the state of the system after the measurement if $+1$ is obtained.

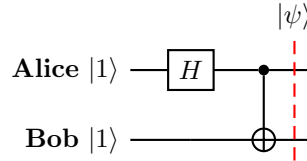
$$\begin{aligned}
\mathbb{P}[+1] &= \left(|0\rangle, \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} \right) |0\rangle \right) \\
&= \left(|0\rangle, \left(\frac{I |0\rangle + (\hat{v} \cdot \hat{\sigma}) |0\rangle}{2} \right) \right) \\
&= \left(|0\rangle, \left(\frac{|0\rangle + v_1 \sigma_1 |0\rangle + v_2 \sigma_2 |0\rangle + v_3 \sigma_3 |0\rangle}{2} \right) \right) \\
&= \left(|0\rangle, \left(\frac{|0\rangle + v_1 |1\rangle + v_2 i |1\rangle + v_3 |0\rangle}{2} \right) \right) \\
&= \left(|0\rangle, \left(\frac{(1 + v_3) |0\rangle + (v_1 + i v_2) |1\rangle}{2} \right) \right) \\
&= \left(|0\rangle, \frac{(1 + v_3) |0\rangle}{2} + \frac{(v_1 + i v_2) |1\rangle}{2} \right) \\
&= \frac{1 + v_3}{2}
\end{aligned}$$

The state of the system post-measurement is

$$\begin{aligned}
\frac{\frac{I + \hat{v} \cdot \hat{\sigma}}{2} |0\rangle}{\sqrt{\mathbb{P}[+1]}} &= \frac{\frac{(1 + v_3) |0\rangle + (v_1 + i v_2) |1\rangle}{2}}{\sqrt{\frac{1 + v_3}{2}}} \\
&= \frac{(1 + v_3) |0\rangle + (v_1 + i v_2) |1\rangle}{\sqrt{2(1 + v_3)}}
\end{aligned}$$

10.1 EPR Experiment

Consider the entangled pair



where $\psi = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$ of qubits belonging to Alice and Bob respectively. Suppose Alice and Bob are a long way away from one another.

Alice performs a measurement using the observable $\hat{v} \cdot \hat{\sigma}$ (for some $\hat{v} \in \mathbb{R}^3$). Suppose Alice receives the result $+1$. Then the post-measurement state at Bob's qubit is computed as follows. First compute,

$$\begin{aligned}
\left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} \otimes I \right) |\psi\rangle &= \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} \otimes I \right) \frac{|01\rangle - |10\rangle}{\sqrt{2}} \\
&= \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} \otimes I \right) \frac{|0\rangle |1\rangle - |1\rangle |0\rangle}{\sqrt{2}} \\
&= \frac{1}{\sqrt{2}} \left[\left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} |0\rangle \otimes I |1\rangle \right) - \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} |1\rangle \otimes I |0\rangle \right) \right] \\
&= \frac{1}{\sqrt{2}} \left[\left(\frac{(1 + v_3) |0\rangle + (v_1 + i v_2) |1\rangle}{2} \right) \otimes |1\rangle - \left(\frac{(v_1 - i v_2) |0\rangle + (1 - v_3) |1\rangle}{2} \right) \otimes |0\rangle \right] \\
&= \frac{1}{2\sqrt{2}} [(1 + v_3) |01\rangle + (v_1 + i v_2) |11\rangle - (v_1 - i v_2) |00\rangle - (1 - v_3) |10\rangle]
\end{aligned}$$

Next compute,

$$\begin{aligned}
\left(|\psi\rangle, \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} \otimes I \right) |\psi\rangle \right) &= \left(\frac{|01\rangle - |10\rangle}{\sqrt{2}}, \frac{1}{2\sqrt{2}} [(1 + v_3) |01\rangle + (v_1 + iv_2) |11\rangle - (v_1 - iv_2) |00\rangle - (1 - v_3) |10\rangle] \right) \\
&= \frac{(1 + v_3)}{4} + \frac{(1 - v_3)}{4} \\
&= \frac{1}{2}
\end{aligned}$$

There, conditioned on Alice's observation result is +1, the Bob's system collapses to

$$\begin{aligned}
\frac{(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} \otimes I) |\psi\rangle}{\sqrt{(|\psi\rangle, (\frac{I + \hat{v} \cdot \hat{\sigma}}{2} \otimes I) |\psi\rangle)}} &= \frac{\frac{1}{2\sqrt{2}} [(1 + v_3) |01\rangle + (v_1 + iv_2) |11\rangle - (v_1 - iv_2) |00\rangle - (1 - v_3) |10\rangle]}{\sqrt{1/2}} \\
&= \frac{1}{2} [(1 + v_3) |01\rangle + (v_1 + iv_2) |11\rangle - (v_1 - iv_2) |00\rangle - (1 - v_3) |10\rangle]
\end{aligned}$$

If now Bob were to perform measurement using the observable $\hat{v} \cdot \hat{\sigma}$ (the same $\hat{v}$ as used by Alice), compute $\mathbb{P}_{\text{Bob}}[+1]$. If

$$|\phi\rangle = \frac{1}{2} [(1 + v_3) |01\rangle + (v_1 + iv_2) |11\rangle - (v_1 - iv_2) |00\rangle - (1 - v_3) |10\rangle]$$

then, $\mathbb{P}_{\text{Bob}}[+1] = \langle \phi |, I \otimes \frac{I + \hat{v} \cdot \hat{\sigma}}{2} | \phi \rangle$.

Let's first compute,

$$\begin{aligned}
I \otimes \frac{I + \hat{v} \cdot \hat{\sigma}}{2} |\phi\rangle &= \left(I \otimes \frac{I + \hat{v} \cdot \hat{\sigma}}{2} \right) \left(\frac{1}{2} [(1 + v_3) |01\rangle + (v_1 + iv_2) |11\rangle - (v_1 - iv_2) |00\rangle - (1 - v_3) |10\rangle] \right) \\
&= \left(I \otimes \frac{I + \hat{v} \cdot \hat{\sigma}}{2} \right) \left(\frac{1}{2} [((1 + v_3) |0\rangle + (v_1 + iv_2) |1\rangle) |1\rangle - ((v_1 - iv_2) |0\rangle + (1 - v_3) |1\rangle) |0\rangle] \right) \\
&= \frac{1}{2} \left[((1 + v_3) |0\rangle + (v_1 + iv_2) |1\rangle) \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} |1\rangle \right) - ((v_1 - iv_2) |0\rangle + (1 - v_3) |1\rangle) \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} |0\rangle \right) \right]
\end{aligned}$$

Check that

$$\begin{aligned}
&((1 + v_3) |0\rangle + (v_1 + iv_2) |1\rangle) \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} |1\rangle \right) \\
&= ((1 + v_3) |0\rangle + (v_1 + iv_2) |1\rangle) \left(\frac{(v_1 - iv_2) |0\rangle + (1 - v_3) |1\rangle}{2} \right) \\
&= \frac{(1 + v_3)(v_1 - iv_2) |00\rangle + (1 - v_3^2) |01\rangle + (v_1^2 + v_2^2) |10\rangle + (v_1 + iv_2)(1 - v_3) |11\rangle}{2}
\end{aligned}$$

Similarly, check that

$$\begin{aligned}
&((v_1 - iv_2) |0\rangle + (1 - v_3) |1\rangle) \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} |0\rangle \right) \\
&= ((v_1 - iv_2) |0\rangle + (1 - v_3) |1\rangle) \left(\frac{(1 + v_3) |0\rangle + (v_1 + iv_2) |1\rangle}{2} \right) \\
&= \frac{(1 + v_3)(v_1 - iv_2) |00\rangle + (v_1^2 + v_2^2) |01\rangle + (1 - v_3^2) |10\rangle + (v_1 + iv_2)(1 - v_3) |11\rangle}{2}
\end{aligned}$$

Therefore,

$$\begin{aligned}
I \otimes \frac{I + \hat{v} \cdot \hat{\sigma}}{2} |\phi\rangle &= \frac{1}{2} \left[((1 + v_3) |0\rangle + (v_1 + iv_2) |1\rangle) \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} \right) |1\rangle - ((v_1 - iv_2) |0\rangle + (1 - v_3) |1\rangle) \left(\frac{I + \hat{v} \cdot \hat{\sigma}}{2} \right) |0\rangle \right] \\
&= \frac{1}{2} \left[\frac{(1 - v_3^2 - v_1^2 - v_2^2) |01\rangle + (v_1^2 + v_2^2 - 1 + v_3^2) |10\rangle}{2} \right] \\
&= 0
\end{aligned}$$

Therefore, $\mathbb{P}_{\text{Bob}}[+1] = (\langle \phi |, I \otimes \frac{I + \hat{v} \cdot \hat{\sigma}}{2} |\phi\rangle) = (\langle \phi |, 0 |\phi\rangle) = 0$. Hence, $\mathbb{P}_{\text{Bob}}[-1] = 1$.

Similarly, it is straightforward to verify that if Alice's measurement outcome is -1 , then $\mathbb{P}_{\text{Bob}}[+1] = 1$.

Therefore, we have established the following: irrespective of the choice of direction $\hat{v}$ made by Alice, the outcome of Alice's and Bob's measurements along $\hat{v}$ are always opposite. In particular,

Alice can predict with certainty the outcome of Bob's measurement along the direction of $\hat{v}$.

In summary, we have prepared an entangled two-qubit system and spatially separated the qubits, with one held by Alice and the other by Bob. For any direction $\hat{v} \in \mathbb{R}^3$, if Alice measures $\hat{v} \cdot \hat{\sigma}$ on her qubit, she can determine with certainty what the outcome of Bob's measurement along the same direction would be.

In the celebrated paper by Albert Einstein, Boris Podolsky, and Nathan Rosen (the EPR paper), they proposed the following criterion: *If it is possible to predict with certainty (i.e., with probability 1) the value of a physical quantity immediately before measurement, and without in any way disturbing the system, then there exists an element of reality corresponding to that physical quantity.*

EPR argued that in the situation described above, the predicted measurement outcome for Bob's system satisfies this criterion and therefore constitutes an element of reality. However, standard quantum mechanics does not provide any fundamental representation of such predetermined values for all possible measurement directions.

They therefore concluded that the quantum-mechanical description of physical reality is incomplete.